

# SERGEY CHERNYAKOV



## AI Engineering Lead · Senior ML / AI Engineer · LLM Integration Architect

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Age: 47 · Male

GitHub: [github.com/sergeychernyakov](https://github.com/sergeychernyakov)

Location: Wrocław, Poland · Remote / EU (open to relocation, hybrid 1–2 days/week OK)

## PROFESSIONAL SUMMARY

Senior engineer with **20+ years** of commercial experience and current focus on **AI / ML engineering, LLM integration, multi-agent systems and AI automation in production**. Designs and operates real-time ML pipelines, OCR / document-understanding systems, retrieval-augmented services, MCP (Model Context Protocol) servers and AI-driven backend services for international clients.

Strong Python and backend foundations: **FastAPI, Django, Flask, PostgreSQL, Docker, CI/CD**. Hands-on with **OpenAI, Anthropic Claude, XGBoost, LangChain / LangGraph, MCP / SSE, Hugging Face**. Long track record of long-term contracts in the US, UK, EU and Australia — Top Rated on Upwork, **18,000+ hours, 71+ contracts**.

Available as **B2B contractor via Deel.com**.

## KEY SKILLS

- **AI / LLM platforms:** OpenAI API (GPT-4o / o1) · Anthropic Claude API · Hugging Face · open-source LLMs
- **AI engineering patterns:** Multi-agent systems · RAG · Vector DBs · Prompt engineering · LangChain / LangGraph · Model Context Protocol (MCP) servers + SSE · AI-assisted SDLC (Claude Code agents — planner / architect / coder / reviewer / QA)
- **ML:** XGBoost · scikit-learn · feature engineering · quantile regression · survival / hazard models · schema-versioned datasets · feature-stability protocols · backtest-live parity
- **Document AI / OCR:** Google OCR / Vision · GPT-Vision · PDF / Word processing · layout-preserving translation · vision-LLMs for image classification + metadata extraction
- **Voice / Conversational AI:** OpenAI Whisper (STT) · TTS · WhatsApp & Telegram Bot APIs · Asterisk telephony integration
- **Backend foundations:** Python (FastAPI / Django / Flask) · async I/O · Docker · GitHub Actions CI/CD
- **Data:** PostgreSQL · MySQL · Redis · Kafka · MongoDB · ETL pipelines · streaming
- **Integrations:** OpenAI · Telegram · WhatsApp · Stripe · PayPal · Salesforce · Twilio · Solidity / Web3.py

## PROFESSIONAL EXPERIENCE

### AI Engineering Lead / AI Automation Architect — Sabermine (United Kingdom, Remote)

November 2024 – November 2025

Led AI engineering and automation direction for internal and client systems.

- Designed and operated Python-based **ETL and automation pipelines** for data ingestion, transformation and downstream analytics
- Built **multi-agent AI systems** for development, QA, documentation and DevOps automation
- Backend services on **Python, FastAPI, PostgreSQL**; Docker-based deployment, CI/CD via GitHub Actions
- Integrated **OpenAI APIs / LLM workflows** into production for OCR, document classification and content generation
- Built internal tooling to improve data quality, observability and reproducibility of AI pipelines
- Improved development speed and operational efficiency **5–10x** via AI-assisted SDLC

### Senior AI / Principal Backend Engineer — Upwork (International Clients, Remote)

June 2012 – June 2025 (Top Rated)

**18,000+ hours · 71+ contracts · 98% Job Success**. Long-term AI engagements (2023–2026):

🔗 **LLM-integrated products**

- **Medical document translation pipelines** (`umetex`) — Django + ChatGPT translation of medical PDFs / Word / images preserving layout (tables, images), Cyrillic font handling, predefined-phrase consistency, IP-based history, async processing.
- **Equipment recognition from photos** (`tagman`) — GPT-Vision-based image classification + structured metadata extraction (serial numbers, equipment labels) with Excel report export.
- **WhatsApp + OpenAI chatbot** (`instreamatic`) — real-time AI responses with MongoDB session persistence (context-aware multi-turn conversations).
- **Multi-modal Telegram bots** (`aihelper`) — Translator (RU/UA + speech) and Diet assistant with text/photo/voice/video/document support, OpenAI threads with auto-refresh, monetization via Telegram payments, i18n (en/ru/ua), TTS.

#### 🔗 Conversational / voice AI

- **OpenAI Voice Assistant on Asterisk telephony** (`asterisk`) — AI assistant handles incoming calls, transcribes speech (Whisper), generates contextual responses (GPT), TTS playback, SMTP summary delivery.

#### 🔗 Real-time ML pipelines

- **Solana Pulse Bot** (`solana_pulse_bot`, archived 2026-05) — observer bot for Solana memecoin launchpads (pump.fun, letsbonk). Two-phase scoring: fast filter at T+5s + full analysis at T+90s. **82 features per token** (schema v21), **8 XGBoost ML heads** (entry main / entry @T+30 / entry regression / entry-timing 3-class / survival hazard / SL+TP+max\_hold quantile heads). **100% backtest-live decision parity** verified on 300+ tokens. Daily 9 honesty checks (shuffled labels, prior drift, economic backtest), 5-seed feature-stability protocol, config-hash drift WARN at runtime, schema versioning. Stack: Python, asyncpg + psycopg2, Streamlit dashboards, Helius API, WebSockets.

#### 🔗 MCP servers + AI agents

- **AQ AI Agent** (`aq_ai_agent`) — FastAPI + Model Context Protocol via Server-Sent Events; exposes legacy Aquarium SOAP CRM as agent-callable MCP tools. Demonstrates pattern for wrapping legacy enterprise APIs into modern AI-agent ecosystems.
- **Weather MCP Agent** (`weather_mcp_agent`) — clean FastAPI + MCP SSE reference implementation, FastAgent client integration.
- **AI-driven SDLC tooling** (`blank_python_project`) — Claude Code agents for automated planner / architect / coder / reviewer / QA workflows; pre-commit (isort/black/ruff/pylint), log rotation, type hints.

#### 🔗 Content automation with AI

- **Pauchok** (private) — automated content publishing to Yandex.Dzen with ChatGPT image generation, RSS-driven scheduling, Playwright browser automation.
- **TG Promoter** (`tgwriterbot`) — Telegram automation with multi-account rotation, anti-ban logic (daily/hourly limits, cooldowns), Claude CLI integration for text generation.
- **ImWriter** (private, Rails 7.2) — AI book constructor: chapter generation, character creation, full reviews, point-by-point text patches, grammar correction.

#### 🔗 NLP & sentiment

- **Auto Sentiment** (`auto_sentiment`) — sentiment analysis on car-dealership customer comments; Excel-based business analytics pipeline.

### Senior Fullstack Engineer — Fit2you (Italy, insurance)

April 2022 – January 2024

- Internal enterprise systems for an Italian insurance company (dealer / vendor / client portals)
- Migrated all company applications from **Rails 4.1** → **Rails 7.1** (5 legacy apps), SonarQube quality gates, 80% test coverage
- Supported and developed **all 8 of the company's applications** end-to-end — feature work, maintenance and incident fixes; one of them was a **Node.js + React** app, the rest on Ruby on Rails
- Stack: Ruby on Rails, Node.js / Express, TypeScript, PostgreSQL

### Earlier roles — Clickfunnels (USA), Lawdingo (USA), Gleam.io (Australia), Skykillers Digital Agency, ITC Publishing, RBC-Ukraine

2004 – 2022

Senior backend engineering for high-traffic media platforms, advertising agency campaigns (Colgate, Danone, Nicorette, Visine, Davidoff, Johnson's Baby), payment platforms (Lawdingo: Stripe + Twilio + Sidekiq), high-volume marketing SaaS (Clickfunnels: Rails + MySQL + Kafka + Redis + 12+ integrations), SaaS competitions platform (Gleam.io: Rails + Angular + social APIs).

Reference from Ryan Montgomery (COO, ClickFunnels): "Seriously one of the best hires we've ever made on Upwork. Sergey has continued to exceed our expectations."

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## SELECTED OPEN-SOURCE / PORTFOLIO PROJECTS

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- **solana\_pulse\_bot** — Real-time ML scoring with 8 XGBoost heads, 100% backtest-live parity. [github.com/sergeychernyakov/solana\\_pulse\\_bot](https://github.com/sergeychernyakov/solana_pulse_bot)
- **umetex** — Django + ChatGPT medical document translator. [github.com/sergeychernyakov/umetex](https://github.com/sergeychernyakov/umetex)
- **aq\_ai\_agent** — FastAPI + MCP + SOAP CRM integration. [github.com/sergeychernyakov/aq\\_ai\\_agent](https://github.com/sergeychernyakov/aq_ai_agent)
- **weather\_mcp\_agent** — Clean MCP SSE reference. [github.com/sergeychernyakov/weather\\_mcp\\_agent](https://github.com/sergeychernyakov/weather_mcp_agent)
- **asterisk** — OpenAI voice AI on Asterisk telephony. [github.com/sergeychernyakov/asterisk](https://github.com/sergeychernyakov/asterisk)
- **aihelper** — Multi-modal Telegram bots with OpenAI + payments. [github.com/sergeychernyakov/aihelper](https://github.com/sergeychernyakov/aihelper)
- **tgwriterbot** — TG automation with Claude CLI + anti-ban. [github.com/sergeychernyakov/tgwriterbot](https://github.com/sergeychernyakov/tgwriterbot)
- **blank\_python\_project** — Python template with Claude Code SDLC agents. [github.com/sergeychernyakov/blank\\_python\\_project](https://github.com/sergeychernyakov/blank_python_project)

**Private:** `arbitraflashbot_v2` (DEX arbitrage with ML + flashloans), `imwriter_wizard` (Rails AI book constructor), `pauchok` (Yandex.Dzen auto-publish), `tagman` (GPT-Vision OCR web app), `instreamatic` (WhatsApp + GPT bot), `frisson` (meter-reading neural net), `surveye` (video analytics).

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## EDUCATION

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**Bachelor's Degree** — Don State Technical University, 1999

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## LANGUAGES

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Ukrainian — Native · English — C1 (Fluent professional) · Polish — Learning (A1)

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## ENGAGEMENT

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- **B2B contractor** — invoicing via **Deel.com** (current setup; open to other platforms)
- **Format:** Remote (preferred) · Hybrid OK (1–2 days/week)
- **Location / mobility:** Wrocław, Poland · Open to relocation within EU · Visa sponsorship welcome for non-Poland EU
- **Notice period:** ~1 month
- Strong production / operational mindset; AI quality, observability and reproducibility as defaults